

NAME: \_\_\_\_\_

PERIOD: \_\_\_\_\_

DATE: \_\_\_\_\_

## Can Chance Make a Streak?

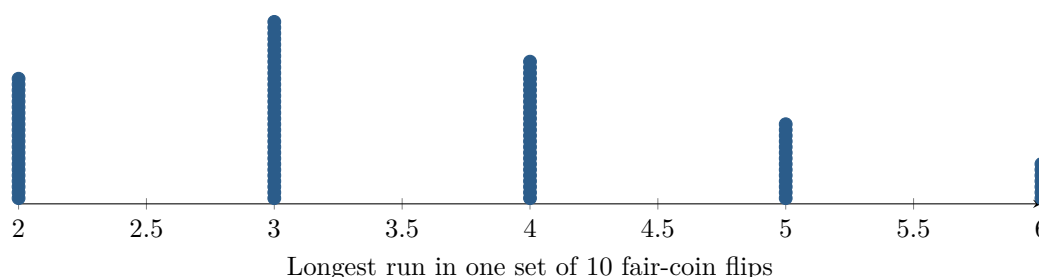
AP Statistics · Topic 4.1 (CED 2.3) · B: 2026-10-19 · E: 2026-11-02

### [STAR] TODAY'S PROBLEM

Suppose a coin advertised as fair is flipped 10 times and gives HHHHHHTTTT. To investigate what chance alone could produce, a class simulates 100 sets of 10 independent fair-coin flips. For each set, the class records the longest run of matching outcomes; the dotplot shows the results.

**Claim A:** “The coin is rigged—six heads in a row cannot be chance.”

**Claim B:** “Streaks like this can happen by chance; simulation can tell us how unusual this one is.”



### FIRST TAKE — before the video (5 min)

Which claim seems better supported right now? Cite one detail from the sequence or dotplot.

### [BOOK] MODEL THE PATTERN, NOT THE STORY (keep on the page)

Random processes can produce visible patterns, including streaks. To judge a pattern under a proposed chance model, make each simulation trial match the original process, choose a statistic that measures how extreme the pattern is, and count results **at least as extreme** as the observation. Repeat many independent trials; the successful-trial proportion estimates how often chance alone would produce such a result. An unusual result is evidence against a model, not proof of what caused the departure.

**Finish after the video:**

**DOK 1** (a) State the longest run in the observed sequence and the most common longest-run value among the simulated sets.

**DOK 2** (b) Use the dotplot to estimate the probability that 10 fair-coin flips have a longest run at least as long as the observed run. Interpret the estimate in context.

**DOK 3** (c) Which claim is better supported by this evidence? Cite the simulation result that settles your choice. Then describe a larger simulation that would more precisely settle the disagreement: state what one trial is, what to record and count, how many trials to use, and what result would make you doubt that the coin is fair.

**Frame:** Claim \_\_\_\_ is better supported because \_\_\_\_\_.

**Frame:** One trial is \_\_\_\_\_; I would record \_\_\_\_\_ and doubt fairness if \_\_\_\_\_.

**EXIT — turn this in**

One thing the video changed about my first take: \_\_\_\_\_